Building an Information Pipeline and Evaluating a Retrieval-Augmented Chatbot for Climate Education in Toronto

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Introduction

With increasing prevalence use of LLM's over classical search engines for information retrieval for detailed responses and surface level research, a chatbot is created and purposed to reduce climate illiteracy and educate Toronto's local communities about issues of climate change, and to suggest actionable plans. Advantages of using a chatbot over traditional education methods is that learners gain a more personalized learning experience, and for educators they gain time-saving assistance and improved pedagogy [1]. However, modern LLMs face issues of accuracy and reliability, and hence the implementation of a retrieval-augmented generated architecture.

Current LLM response face issues of "hallucinations" during the model's reasoning processes or a lack of relevancy to inputted prompts [2]. RAG architecture reduces the probability of these events by automatically detecting documents within a static database relevant to the user's prompt, attaching it to the prompt, then sends it to the LLM to be processed before generating a response, increasing specificity of the query within the LLM to generate more accurate and focused responses (see figure a). However, it remains a discussion whether RAG implementation raises a significant improvement in generated response for climate change related discussions. Using a multilingual chatbot could overcome issues in climate justice and environmental conservation [3].

The research is divided into two segments: 1) Computationally constructing a data pipeline for the RAG's architecture to provide up-to-date information to its static database and 2) evaluating the responses of a chatbot before and after documents from the data pipeline and integrating it to the chatbot's systems and evaluating the suitability of a retrieval-augmented generation (RAG) architecture for integration within educational contexts to have better human alignment. A rating criteria for humans to grade chatbot responses was developed based on a study on natural-language generated responses [4].

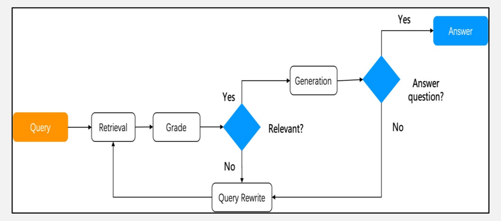

Figure a – RAG flow process for LLMs. From RAGFlow.io

Methods

Data pipeline construction:

A database (DB) was constructed to log pre-existing documents and serve as a quick check for any new documents that are scrapped. Documents are pipelined to GeminiAPI to scrap information about the title of the documents, authors, published date, tags, DOI/URL sourced from if available.

Another API was made with Elsevier to source academic documents, which gets scholarly articles via a keyword-based approach. Similarly, a web scraper was developed to source documents by determining good starting URLs and completing both a depth search and site map crawl via keywords to return relevant documents. Newly source documents are assigned a relevancy score to climate change, which requires a relevancy index.

A relevancy index was constructed to automatically determine the practicality of the sourced information for the chatbot and to automate data uptake processes, by determining how well direct quoted text from the documents can respond to a set of given queries via NLP methods. This was done by chunking the documents via Sentence Transformers and checking with cosine similarity (using Cosine Similarity library). These queries are aimed at asking questions about scientific facts about climate change, misinformation about climate change, climate change related mental health issues, and more. A control group of documents was fed to the relevancy index to determine a baseline score. After sourcing the documents, we are also able to generate an atlas by partnering with Nomic AI to generate keywords within our database, identifying common topics and information gaps within our system.

Chatbot response evaluations:

A set of queries aimed at discussing how climate change might impact a member of the general public's daily routine physically, socio-economically and politically was created. The responses were evaluated computationally and by user participants. Two different set of queries were generated – One set more focused on scientific accuracy, ongoing socio-political issues, revolving around topics of discussions in higher education, and another set of queries more probable to be asked by ordinary citizens regarding situations that may affect their everyday life (e.g. how might climate change affect their commute, bills, finances, health, seniors, mental health). Participants were exposed to responses by a chatbot without being fed documents for RAG and with.

To evaluate the responses computationally, it was weighted based on scientific accuracy (25%), misinformation-handling (20%), climate psychology (mental health, climate anxiety related discussions, 15%), uncertainty communication/confidence score (10%), source-hierarchy (how reliable or reputable a source is, 15%), and solution-framing/actionability of the response (15%). This returns a score between 0 and 1 for each response, and the final score is taken by averaging the score over all responses.

To score the responses by humans, a grading criteria of 1-5 was developed, and participant(s) were asked to score the chatbot responses considering the following factors: Accuracy and faithfulness, relevancy and local grounding (to Toronto), completeness and specificity (Did it go beyond "high-level" advice to spell out concrete steps, timelines, or numbers? Are there clear examples ("here's the rebate program name," "here's the TTC contingency routing," etc.) rather than just "you could apply for grants"?), actionability (can someone walk away and do something?), and clarity and readability, based on . The final score is taken as the average over all responses (see figure 7). A generic surface level survey was also sent out, asking anonymous participants for their response preference between the non-RAG and RAG response without rating responses (see figure 6).

To replicate testability of generated responses, an open source LLM and embedding model was used (llama2:7b and sentence-transformers all-MiniLM-L7-v2), and the chatbot itself is available on GitHub.

Results

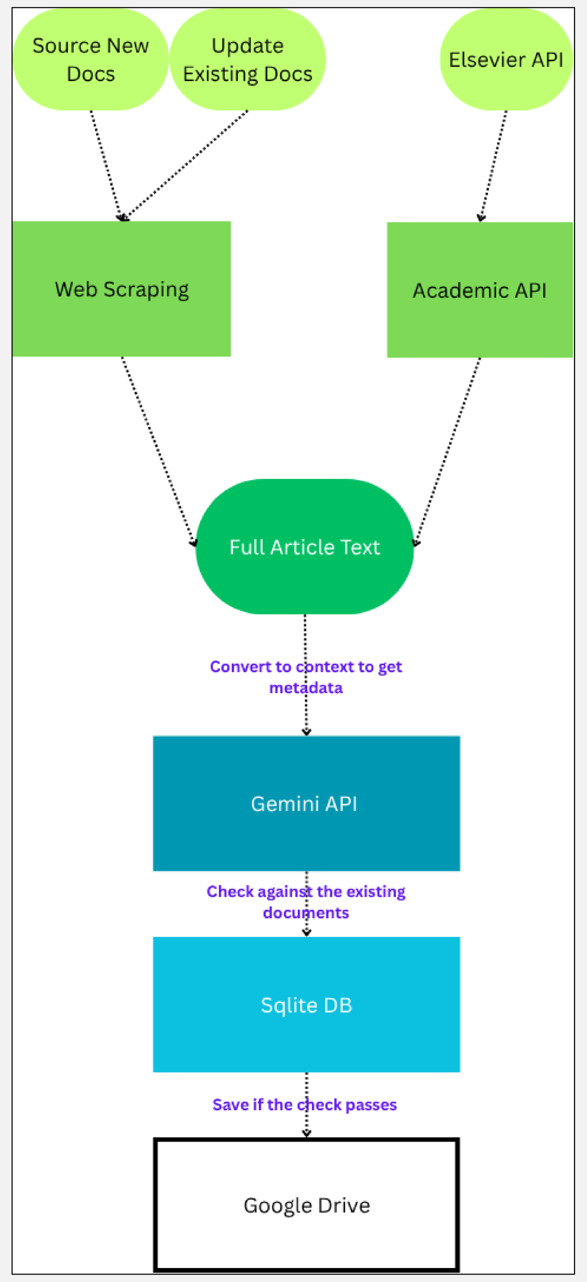

Figure 1: General data pipeline workflow process

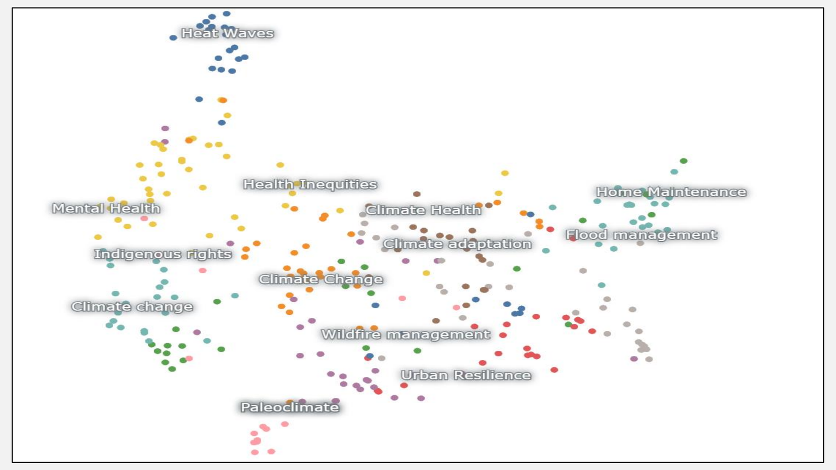

Figure 2: Keyword map of documents (pre-pipeline) to identify information gaps and existing topics, generated with Nomic.

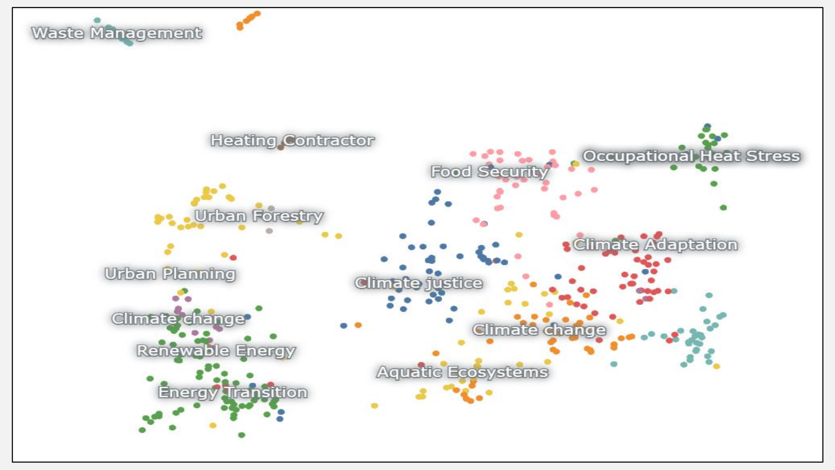

Figure 3: Keyword map of documents (pipelined) to identify information gaps and existing topics, also generated with Nomic.

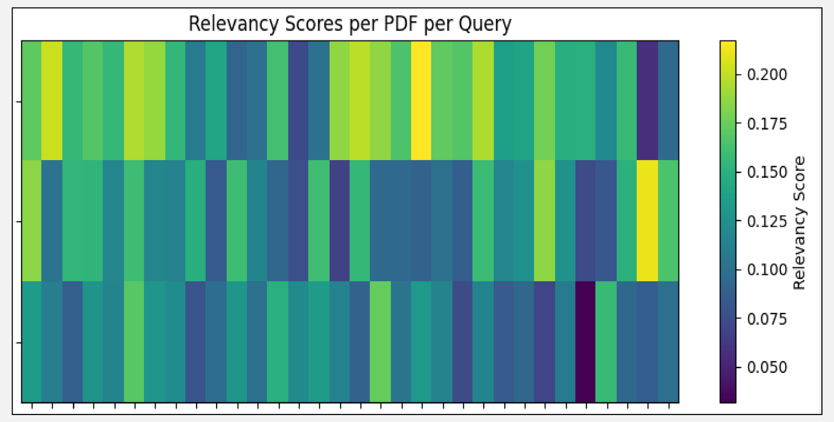

Figure 4: Relevancy index testing on three documents as an example, each column representing a query and each row being a document.

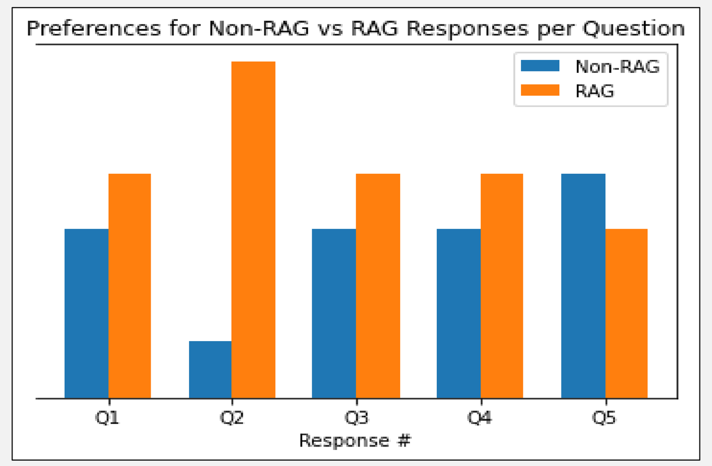

Figure 6: Participant preference for responses of the bot with RAG implementation and without for the first five queries given to it.

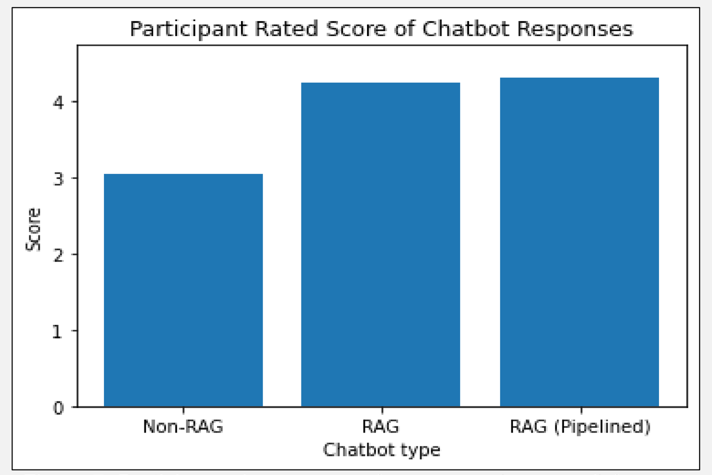

Figure 7: Participant given scores of responses (averaged) from different versions of the chatbot based on a 1-5 grading criteria.

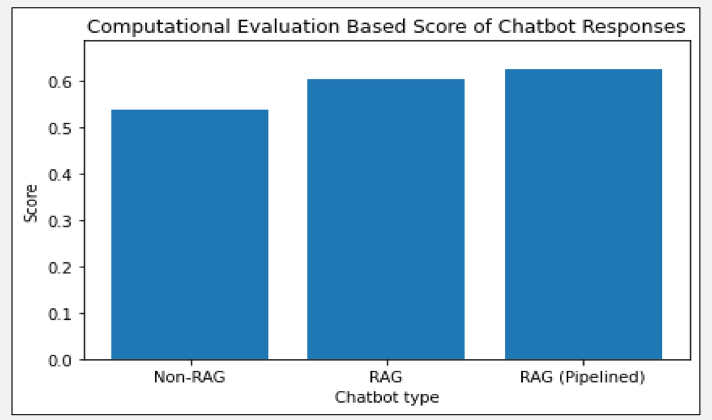

Figure 8: Computationally evaluated scores of responses (averaged) from different versions of the chatbot on a 0-1 criteria.

Recommendations

- **Technical Improvements**: The information pipeline in its current stage is a proof of concept with different sections working separately. The first improvement that can be made it to integrate the relevancy index into the DB that can automatically discard documents that are not relevant to the chatbot. To support continuous and up-to-date data ingestion, we also recommend integrating GitHub Actions into the pipeline to automate document retrieval and processing on a scheduled basis. Additionally, the current database structure - centered around a single documents table - is not optimal for scalability. A more modular schema with separate tables for metadata, keyword mappings, relevancy scores, and source tracking would enhance query flexibility, indexing efficiency, and future expansion (e.g., user feedback or version history tracking). Future development should focus on tighter integration between modules, allowing for real-time indexing, scoring, and updating of documents to support a scalable and adaptive RAG system.
- **Practical and Educational Integration:** To maximize community impact and support climate education goals, we recommend deploying the chatbot through partnerships with educators, city agencies, and local NGOs. It can function as a public-facing tool for climate literacy, policy awareness, and behavior change, offering accurate and actionable information in real-time. Integrating the chatbot into libraries, community centers, classrooms, and outreach programs would provide residents with an accessible way to engage with complex climate issues. From an educational standpoint, the chatbot supports personalized, inquiry-based learning, while for policymakers and civic institutions, it serves as a scalable platform for disseminating local climate programs, rebate schemes, emergency protocols, and sustainability initiatives. By embedding the chatbot into Toronto's public infrastructure, it becomes a bridge between policy, pedagogy, and public engagement.
- **Evaluation and Future Research:** To ensure long-term effectiveness and public trust, we recommend expanding the evaluation framework and investing in iterative testing across diverse and larger user groups, reflecting Toronto's multicultural and multilingual population. The chatbot should be tested not only for scientific accuracy but also for cultural sensitivity, language accessibility, and relevance to a wide range of lived experiences. The existing evaluation process - combining computational scoring with human feedback - can be extended with longitudinal studies that assess knowledge retention, behavioral change, and community impact. Future work should explore multilingual support, bias detection, and personalized response strategies to better serve different communities. These enhancements will help ensure that the chatbot is inclusive, equitable, and representative of Toronto's diverse climate concerns. Furthermore, in evaluation grading by humans, different sets of rubrics should be developed to account for possible variances on how chatbot responses are judged.

Conclusions

From a technical standpoint, this project demonstrates the feasibility of building a RAG-enabled climate education chatbot using open-source tools and modular APIs. While the pipeline is still a proof of concept, it effectively combines document scraping, embedding, and retrieval workflows, including the use of the Elsevier API for academic content. A relevancy index was developed to assess document utility, but it is not yet integrated into the pipeline for automated filtering, which remains a key area for future improvement. The current single-table database structure also limits scalability, underscoring the need for a more modular schema to support richer querying and metadata tracking. Although Nomic's semantic visualizations were visually appealing, they proved limited in analytical value and practical guidance for database optimization.

Whilst ratings for the RAG responses are typically higher, we see no statistical significance in terms of which response users prefer in the survey when carrying out a binary test on a 50-50 probability level. For responses given by humans, they often noted that RAG aided responses were more specific and actionable, whereas non-RAG responses provided generic level response and surface level action such as "You can help reduce flooding by reducing your carbon footprint...".

Whilst quantitative improvements in RAG responses can be observed by both computational algorithms and human criteria in this project, it appears that there are limits to improvements in responses made via RAG, where continuously increasing the number of relevant documents sourced to its static database yields no significant benefits (apparent in both responses scored by participants and computational evaluations). Whilst the findings in this project suggests that RAG is a valid method to optimize LLM inference for better human alignment in climate chatbots, it also leaves the door open for other methods for NLG optimization.

Literature cited

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Further information

Feel free to reach out to Climate Resilience Communities for the Chatbot, or members of the pipeline and evaluation project via email.

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